Agios Pharmaceuticals, Inc. v. Annora Pharma Private Limited et al

Plaintiff:	Agios Pharmaceuticals, Inc.
Defendant:	Annora Pharma Private Limited, Hetero Labs Limited and Hetero USA Inc.
Case Number:	1:2026cv00603
Filed:	May 26, 2026
Court:	U.S. District Court for the District of Delaware
Presiding Judge:	Colm F Connolly
Nature of Suit:	Patent - Abbreviated New Drug Application(ANDA)
Cause of Action:	35 U.S.C. § 271 Patent Infringement
Jury Demanded By:	None

Date Filed	Document Text
July 14, 2026	Pro Hac Vice Attorney Neal Seth for Annora Pharma Private Limited, Hetero Labs Limited & Hetero USA Inc. added for electronic noticing. Pursuant to Local Rule 83.5 (d)., Delaware counsel shall be the registered users of CM/ECF and shall be required to file all papers. (slk)
July 13, 2026	Pro Hac Vice Attorney Teresa Summers for Annora Pharma Private Limited, Hetero Labs Limited,and Hetero USA Inc. added for electronic noticing. Pursuant to Local Rule 83.5 (d)., Delaware counsel shall be the registered users of CM/ECF and shall be required to file all papers. (oam)
July 13, 2026	Pro Hac Vice Attorney Madyson Kent for Annora Pharma Private Limited, Hetero Labs Limited, and Hetero USA Inc. added for electronic noticing. Pursuant to Local Rule 83.5 (d)., Delaware counsel shall be the registered users of CM/ECF and shall be required to file all papers. (oam)
July 9, 2026	SO ORDERED, re #12 MOTION for Pro Hac Vice Appearance of Attorney Neal Seth, Teresa Summers, and Madyson Kent, filed by Hetero USA Inc., Hetero Labs Limited, Annora Pharma Private Limited. Ordered by Judge Colm F. Connolly on 7/9/2026. (kmd)
July 8, 2026	**Filing 13** Pro Hac Vice Fee - Credit Card Payment received for Neal Seth, Teresa Summers, and Madyson Kent. (re #12 MOTION for Pro Hac Vice Appearance of Attorney Neal Seth, Teresa Summers, and Madyson Kent)(Payment of $ 150, receipt

Date Filed	Document Text
	number ADEDC-5011613).(Stamoulis, Stamatios)
July 8, 2026	**Filing 12** MOTION for Pro Hac Vice Appearance of Attorney Neal Seth, Teresa Summers, and Madyson Kent - filed by Annora Pharma Private Limited, Hetero Labs Limited, Hetero USA Inc.. (Stamoulis, Stamatios)
July 6, 2026	**Filing 11** NOTICE of Appearance by Stamatios Stamoulis on behalf of Annora Pharma Private Limited, Hetero Labs Limited, Hetero USA Inc. (Stamoulis, Stamatios)
June 5, 2026	**Filing 10** WAIVER OF SERVICE returned executed by Agios Pharmaceuticals, Inc.: For Hetero USA Inc. waiver sent on 5/29/2026, answer due 8/27/2026. (Tigan, Jeremy)
June 5, 2026	**Filing 9** WAIVER OF SERVICE returned executed by Agios Pharmaceuticals, Inc.: For Hetero Labs Limited waiver sent on 5/29/2026, answer due 8/27/2026. (Tigan, Jeremy)
June 5, 2026	**Filing 8** WAIVER OF SERVICE returned executed by Agios Pharmaceuticals, Inc.: For Annora Pharma Private Limited waiver sent on 5/29/2026, answer due 8/27/2026. (Tigan, Jeremy)
June 3, 2026	**Filing 7** JUDGE COLM F. CONNOLLY STANDING ORDERS. Parties shall comply with all applicable standing orders and consult the Court's website for updates. (mws)
June 3, 2026	Case Assigned to Judge Colm F. Connolly. Please include the initials of the Judge (CFC) after the case number on all documents filed. (alb)
May 27, 2026	**Filing 6** Summons Issued as to Annora Pharma Private Limited on 5/27/2026; Hetero Labs Limited on 5/27/2026; Hetero USA Inc. on 5/27/2026. (Attachments: #1 Summons Issued, #2 Summons Issued)(oam)
May 26, 2026	**Filing 5** Disclosure Statement pursuant to Rule 7.1: No Parents or Affiliates Listed filed by Agios Pharmaceuticals, Inc.. (oam)
May 26, 2026	**Filing 4** Report to the Commissioner of Patents and Trademarks for Patent/Trademark Number(s) 11,234,976 B2; 11,254,652 B2; 11,878,049 B1. (oam)
May 26, 2026	**Filing 3** Supplemental information for patent cases involving an Abbreviated New Drug Application (ANDA) - Date Patentee(s) Received Notice: 04/17/2026. Date of Expiration of Patent: See Attached.Thirty Month Stay Deadline: 8/17/2029. (oam)
May 26, 2026	**Filing 2** Notice, Consent and Referral forms re: U.S. Magistrate Judge jurisdiction. (oam)

Date Filed	Document Text
May 26, 2026	**Filing 1** COMPLAINT against Annora Pharma Private Limited, Hetero Labs Limited, Hetero USA Inc. (Filing fee $ 405, receipt number ADEDC-4977401) - filed by Agios Pharmaceuticals, Inc.. (Attachments: #1 Exhibits A-C, #2 Civil Cover Sheet)(oam)

I. Executive Summary and Procedural Information

- Parties & Counsel:
  - Plaintiff: Agios Pharmaceuticals, Inc. (Delaware)
  - Defendant: Annora Pharma Private Limited (India); Hetero Labs Limited (India); and Hetero USA Inc. (Delaware)
  - Plaintiff's Counsel: Morris, Nichols, Arsht & Tunnell LLP
- Case Identification: 1:26-cv-00603, D. Del., 05/26/2026
- Venue Allegations: Venue is alleged to be proper as to Defendant Hetero USA Inc. because it is a Delaware corporation and therefore resides in the district. Venue is alleged as to the foreign defendants, Annora Pharma Private Limited and Hetero Labs Limited, on the basis that they are foreign corporations that may be sued in any judicial district.
- Core Dispute: Plaintiff alleges that Defendants' filing of an Abbreviated New Drug Application (ANDA) for a generic version of Plaintiff's Pyrukynd® (mitapivat) drug product constitutes an act of infringement of three U.S. patents.
- Technical Context: The lawsuit is in the field of pharmaceutical science, concerning a small-molecule activator of the pyruvate kinase (PK) enzyme for the treatment of hemolytic anemia associated with PK deficiency.
- Key Procedural History: This action arises under the Hatch-Waxman Act following Defendants' submission of ANDA No. 221270, seeking FDA approval to market a generic version of Pyrukynd®. Plaintiff was notified of the ANDA filing, which included a Paragraph IV certification challenging the asserted patents, via a notice letter dated April 16, 2026. This complaint was filed within the 45-day statutory window, triggering a potential 30-month stay of FDA approval for the ANDA.

Case Timeline

Date	Event
2015-06-11	'976 Patent Priority Date
2017-11-22	'652 Patent Priority Date

2019-06-14	'049 Patent Priority Date
2022-02-01	U.S. Patent No. 11,234,976 Issues
2022-02-22	U.S. Patent No. 11,254,652 Issues
2024-01-23	U.S. Patent No. 11,878,049 Issues
2026-04-16	Defendants send Paragraph IV Notice Letter
2026-05-26	Complaint Filed

II. Technology and Patent(s)-in-Suit Analysis

U.S. Patent No. 11,234,976 - "Methods of Using Pyruvate Kinase Activators" (02/01/2022)

The Invention Explained

- Problem Addressed: The patent describes pyruvate kinase deficiency (PKD), a genetic disorder causing an enzyme defect in red blood cells that leads to chronic non-spherocytic hemolytic anemia (HNSHA) '976 Patent, col. 1:15-25 This deficiency results in insufficient levels of adenosine triphosphate (ATP), which is critical for maintaining red blood cell integrity, leading to premature cell destruction and anemia '976 Patent, col. 1:30-49
- The Patented Solution: The invention provides a method for treating PKD by orally administering a specific compound, N-(4-(4-(cyclopropylmethyl)piperazine-1-carbonyl)phenyl)quinoline-8-sulfonamide ("Compound 1"), which acts as an allosteric activator of the pyruvate kinase R (PKR) enzyme '976 Patent, abstract '976 Patent, col. 7:17-30 The claimed method specifies particular oral dosage ranges and frequencies to manage the condition '976 Patent, claim 1
- Technical Importance: The invention provides a potential disease-modifying therapy for a rare disorder for which treatment options were previously limited to palliative care, such as blood transfusions and splenectomy '976 Patent, col. 2:33-39

Key Claims at a Glance

- The complaint asserts at least independent claim 1 Compl. ¶40
- Claim 1 recites:
    - A method for treating pyruvate kinase deficiency (PKD) in a subject.

- The method comprises orally administering to the subject once or twice daily a dose of about 10 mg to about 60 mg, or about 60 mg to about 200 mg.
- The dose is of Compound 1 (or a salt/hydrate thereof), or a pharmaceutical composition containing it.
- Compound 1 is defined as N-(4-(4-(cyclopropylmethyl)piperazine-1-carbonyl)phenyl)quinoline-8-sulfonamide.

U.S. Patent No. 11,254,652 - "Crystalline Forms of N-(4-(4-(Cyclopropylmethyl)piperazine-1-carbonyl)phenyl)quinoline-8-sulfonamide" (02/22/2022)

The Invention Explained

- Problem Addressed: The development of small-molecule drugs requires stable and consistent physical forms of the active pharmaceutical ingredient. The patent notes a need to develop alternative forms of the mitapivat compound to facilitate manufacturing, formulation, and storage stability '652 Patent, col. 1:62-68
- The Patented Solution: The patent discloses specific crystalline salt forms of mitapivat, including a hemisulfate salt designated "crystalline form A" '652 Patent, abstract This form is defined by a unique pattern of peaks in an X-ray powder diffraction (XRPD) analysis, which provides a distinct and reproducible solid-state structure '652 Patent, col. 2:11-17 '652 Patent, claim 1
- Technical Importance: A well-characterized, stable crystalline form is essential for ensuring consistent quality, bioavailability, and shelf-life in a commercial pharmaceutical product.

Key Claims at a Glance

- The complaint asserts at least independent claim 1 Compl. ¶54
- Claim 1 recites:
    - A crystalline form of a compound having a specific chemical structure (mitapivat hemisulfate).
    - The crystalline form is specified as "crystalline form A."
    - Crystalline form A is characterized by x-ray powder diffraction peaks at specific 2θ angles (±0.2°): 9.9°, 15.8°, and 22.6°.

U.S. Patent No. 11,878,049 - "Mitapivat Therapy and Modulators of Cytochrome P450" (01/23/2024)

- Technology Synopsis: The patent addresses the fact that mitapivat is primarily metabolized by the cytochrome P450 3A4/5 (CYP3A4/5) enzymes '049 Patent, col. 1:5-9

This creates a risk for drug-drug interactions when co-administered with other drugs that induce or inhibit these enzymes. The invention provides a method for treating hemolytic anemia by administering mitapivat in combination with a moderate inducer of CYP3A4/5, specifying particular mitapivat dosage regimens for such a scenario '049 Patent, claim 1

- Asserted Claims: The complaint asserts at least independent claim 1 Compl. ¶67
- Accused Features: The infringement allegation is based on the use of Annora's Proposed ANDA Product as will be directed by its proposed label, which is alleged to cover the administration of mitapivat with a moderate inducer of CYP3A4/5 Compl. ¶¶67-69

III. The Accused Instrumentality

Product Identification

"Annora's Proposed ANDA Product," which is a generic version of Pyrukynd® (mitapivat) submitted for FDA approval under ANDA No. 221270 Compl. ¶1

Functionality and Market Context

The accused product is a set of tablets in 5, 20, and 50 mg strengths containing mitapivat (present as mitapivat sulfate) as the active ingredient Compl. ¶1 Compl. ¶36 It is intended to be a bioequivalent substitute for the brand-name drug Pyrukynd® and will be marketed for the same indication: treatment of hemolytic anemia in adults with pyruvate kinase (PK) deficiency Compl. ¶1 The complaint alleges that the product's proposed label will instruct healthcare professionals to prescribe and administer the drug in a manner that infringes the asserted patents Compl. ¶38

IV. Analysis of Infringement Allegations

11,234,976 Infringement Allegations

Claim Element (from Independent Claim 1)	Alleged Infringing Functionality	Complaint Citation	Patent Citation
A method for treating pyruvate kinase deficiency (PKD) in a	Defendants' proposed product is intended for treating PKD, and the	¶40; ¶42	col. 1:15-18

subject in need thereof,	proposed label will instruct its use for this purpose Compl. ¶1		
comprising orally administering to the subject once or twice daily a dose of about 10 mg to about 60 mg or about 60 mg to about 200 mg	The proposed product is supplied in tablet form for oral administration, and the proposed label will allegedly instruct once or twice daily dosing that falls within the claimed ranges Compl. ¶1 Compl. ¶40	¶40; ¶42	col. 7:17-20
of (1) Compound 1 or a pharmaceutically acceptable salt or hydrate thereof... wherein Compound 1 is N-(4-(4-(cyclopropylmethyl)piperazine-1-carbonyl)phenyl)quinoline-8-sulfonamide.	The active ingredient in Defendants' proposed product is mitapivat (Compound 1), present as mitapivat sulfate, which is a pharmaceutically acceptable salt Compl. ¶36 Compl. ¶40	¶36; ¶40	col. 7:21-26

- Identified Points of Contention:
  - Scope Questions: A central question will be one of induced infringement: does the language of the Defendants' proposed product label instruct or encourage physicians to administer the drug in a way that falls squarely within the dosage and frequency limitations of claim 1? The analysis may focus on whether the label mandates, or merely permits, an infringing use.
  - Technical Questions: The court may need to consider whether the "about" language in the dosage claims encompasses the specific dosing regimens that will be recommended in the Defendants' final approved label.

11,254,652 Infringement Allegations

Claim Element (from Independent Claim 1)	Alleged Infringing Functionality	Complaint Citation	Patent Citation
A crystalline form of a compound having the formula: [structure provided]	The complaint alleges that the active ingredient in the proposed ANDA product is mitapivat, present as mitapivat sulfate, which corresponds to the chemical structure depicted Compl. ¶36 Compl. ¶54 The complaint provides a chemical structure of the hemisulfate salt form of the compound claimed in the '652 Patent Compl. ¶54	¶36; ¶54	col. 10:5-25
wherein the crystalline form is crystalline form A characterized by x-ray powder diffraction peaks at 2θ angles (±0.2°) 9.9°, 15.8°, and 22.6°.	On "information and belief," the complaint alleges that the proposed ANDA product contains or is a crystalline salt form covered by at least claim 1 Compl. ¶55	¶55	col. 9:30-33

- Identified Points of Contention:
    - Scope Questions: Practitioners may focus on whether the term "characterized by" requires the presence of only the three listed XRPD peaks, or if it implies a substantial similarity to the overall XRPD pattern of Form A as disclosed in the patent.

- Technical Questions: A primary factual dispute will be whether the Defendants' generic product is, in fact, "crystalline form A." The complaint alleges this on "information and belief," but resolution will depend on expert analysis of the defendants' actual drug substance, likely through techniques like XRPD, once that evidence is produced in discovery.

V. Key Claim Terms for Construction

For U.S. Patent No. 11,234,976:

- The Term: "about" (in the context of dosage, e.g., "about 10 mg to about 60 mg")
- Context and Importance: The definition of "about" is critical for determining the literal scope of the claimed dosage ranges. Defendants may argue their proposed labeled dosing falls outside the claimed ranges, making the interpretation of this term central to the infringement analysis.
- Intrinsic Evidence for Interpretation:
  - Evidence for a Broader Interpretation: The patent uses "about" consistently when reciting dosage ranges in the claims and the specification, suggesting the patentee intended to cover minor, insubstantial variations from the stated numerical values '976 Patent, claim 1 '976 Patent, col. 7:17-20
  - Evidence for a Narrower Interpretation: The patent discloses clinical trial data with specific dose cohorts (e.g., 30 mg, 120 mg, 360 mg) '976 Patent, Figs. 5-6 A party could argue that "about" should be interpreted narrowly, limited to the range of variability expected in standard clinical or manufacturing practice and not extending to entirely different dosing paradigms.

For U.S. Patent No. 11,254,652:

- The Term: "characterized by x-ray powder diffraction peaks at 2θ angles (±0.2°) 9.9°, 15.8°, and 22.6°"
- Context and Importance: This term defines the patented crystalline form. The core of the infringement case for this patent will turn on whether the defendant's product meets this definition. Practitioners may focus on this term because the dispute is likely to be a "battle of the experts" over interpreting analytical data.
- Intrinsic Evidence for Interpretation:
  - Evidence for a Broader Interpretation: The claim language uses "characterized by," not "consisting of" or "having peaks only at." This may support an interpretation that the presence of these three peaks is sufficient for infringement, even if the accused product's XRPD pattern contains additional or different peaks '652 Patent, claim 1

- Evidence for a Narrower Interpretation: The specification provides a more extensive list of 2θ peaks for Form A '652 Patent, col. 9:46-54 and a full XRPD diffractogram '652 Patent, Fig. 1 A party could argue that to be "characterized by" this pattern, an accused product must exhibit a substantial correspondence to this more complete "fingerprint," not just three selected peaks.

VI. Other Allegations

- Indirect Infringement: The complaint alleges Defendants will induce infringement of the method claims of the '976 and '049 patents. The basis for inducement is the allegation that Defendants' proposed product label will instruct and encourage healthcare professionals to administer the drug in a manner that directly infringes the claimed methods of use Compl. ¶49 Compl. ¶75 The complaint also alleges contributory infringement, stating the proposed product is especially made for the patented uses and is not suitable for substantial non-infringing use Compl. ¶50 Compl. ¶76
- Willful Infringement: The complaint does not contain an explicit count for willful infringement. However, it alleges facts that may support a future claim for enhanced damages by asserting that Defendants had pre-suit knowledge of the asserted patents because they are listed in the FDA's Orange Book for Pyrukynd® Compl. ¶48 Compl. ¶61 Compl. ¶74 The complaint also notes that Defendants filed a Paragraph IV certification, acknowledging the patents while asserting their invalidity or non-infringement Compl. ¶34

VII. Analyst's Conclusion: Key Questions for the Case

- A core issue will be one of polymorphic identity: What will the evidence, once produced in discovery, show about the solid-state structure of the Defendants' mitapivat sulfate? Does it exhibit the specific X-ray diffraction peaks that characterize "crystalline form A" as claimed in the '652 patent, or does it constitute a distinct, non-infringing polymorph?
- A second key question will be one of label-driven inducement: For the method-of-use claims in the '976 and '049 patents, does the language of the Defendants' proposed product label actively instruct or encourage physicians to prescribe the generic drug in a way that directly practices every step of the claimed methods, including the specific dosage, frequency, and co-administration limitations?
- A third question concerns claim construction: How broadly will the court interpret the term "about" in the context of the dosage claims of the '976 patent? The outcome of this construction could be determinative of whether the dosing instructions on the accused product's label fall within the literal scope of the claims.

IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF DELAWARE

AGIOS PHARMACEUTICAL, INC.,	)	
Plaintiffs,	)	
	)	
v.	)	C.A. No._____
	)	
ANNORA PHARMA PRIVATE LIMITED,	)	
HETERO LABS LIMITED, AND HETERO	)	
USA INC.,	)	
Defendants.	)	

COMPLAINT

Plaintiff Agios Pharmaceuticals, Inc. ("Agios") by its attorneys, hereby alleges as follows:

NATURE OF THE ACTION

1. This is an action for patent infringement under the patent laws of the United States, Title 35, United States Code, that arises out of the submission by Defendants Annora Pharma Private Limited ("Annora"), Hetero Labs Limited ("Hetero Labs"), and Hetero USA Inc. ("Hetero USA") (collectively "Defendants") of Abbreviated New Drug Application ("ANDA") No. 221270 to the U.S. Food and Drug Administration ("FDA") seeking approval to engage in the commercial manufacture, use, offer for sale, and sale of a generic version of Pyrukynd® (mitapivat) in 5, 20, and 50 mg strength tablets ("Annora's Proposed ANDA Product") prior to the expiration of U.S. Patent Nos. 11,234,976 ("the '976 Patent"); 11,254,652 ("the '652 Patent"); and 11,878,049 ("the '049 Patent") (collectively, "the Asserted Patents"). Defendants notified Agios that they had submitted ANDA No. 221270 by a letter dated April 16, 2026 (the "Notice Letter"). Upon information and belief, if approved by the FDA, Annora's Proposed ANDA Product will be marketed as a competing product to Pyrukynd®, a product developed and marketed by Agios for the treatment of hemolytic anemia in adults with pyruvate kinase (PK) deficiency.

PARTIES

2. Agios is a corporation organized and existing under the laws of the State of Delaware, having a principal place of business at 88 Sidney Street, Cambridge, Massachusetts 02139-4169.
3. On information and belief, Defendant Annora is a corporation organized and existing under the laws of India, having its principal place of business at Sy. No. 261, Plot No. 13 to 14, Annaram Village, Gummadidala Mandal, Sangareddy District, Telangana State, 502313, India.

4. On information and belief, Defendant Hetero Labs is a corporation organized and existing under the laws of India, having a principal place of business at #7-2-A2, Industrial Estates, Sanath Nagar Hyderabad - 500018, Telangana, India.
5. On information and belief, Defendant Hetero USA is a corporation organized and existing under the laws of the State of Delaware (File No. 4837317), having a principal place of business at 1035 Centennial Avenue, Piscataway, New Jersey 08854.
6. On information and belief, Hetero Labs is the parent company of Annora and Hetero USA. See, e.g., *Mirum Pharmaceuticals, Inc., et al., v. Annora Pharma Private Limited, et al.*, C.A. No. 25-1537-MN, D.I. 14 at 5, ¶ 11 (D. Del. Feb. 20, 2026). On information and belief, Annora is a wholly owned subsidiary of Hetero Labs. See, e.g., *id.*
7. On information and belief, Hetero Labs ultimately owns all of Annora's ANDAs, including ANDA No. 221270.
8. On information and belief, Hetero USA is the regulatory agent for Annora and Hetero Labs with respect to ANDA No. 221270. The Notice Letter Agios received from Annora was signed by "Somaraju Indukuri," as the "U.S. Agent for Annora Pharma Private Limited," and represented the same individual as "authorized to accept service of process for any patent infringement complaint" against Annora. On information and belief, Somaraju Indukuri is the Vice President, Regulatory Affairs, U.S. Agent, for Hetero USA. The letter packaging for the Notice Letter identified the sender as "Hetero USA" and the address of the sender as "1035 Centennial Ave[,] Piscataway[,] NJ 08854."
9. On information and belief, Defendants collaborate with respect to, among other things, the development, regulatory approval, marketing, sale, and/or distribution of generic copies of branded pharmaceutical products throughout the United States, including in the State of Delaware. For example, Defendants have admitted that Hetero Labs ultimately owns all of Annora's ANDAs. See, e.g., *Harmony Biosciences, LLC v. AET Pharma US, Inc.*, C.A. No. 23-1340-MN, D.I. 38 at 6, ¶ 21 (D. Del. Mar. 4, 2024). Hetero USA, the agent for Annora, has stated that it is the U.S. sales and marketing arm of Hetero Labs. See *Presence*, Hetero, https://www.hetero.com/presence (last visited May 26, 2026) (listing Hetero USA as "the sales and marketing arm of Hetero [Labs'] Active Pharmaceutical Ingredients (API) and Custom Pharmaceutical Services (CPS) business in USA"). On further information and belief, Defendants are agents of one another and/or operate in concert as integrated parts of the same business group, including with respect to the preparation and submission of ANDA No. 221270 and the development, manufacturing, marketing, sale, and distribution of Annora's Proposed ANDA Product.
10. On information and belief, Hetero Labs and Hetero USA were, are, and will be actively involved in the preparation, submission, maintenance, and further prosecution of ANDA No. 221270, together with Annora, and they intend to directly or indirectly benefit from the preparation, submission, maintenance, and further prosecution of ANDA No. 221270, such that Hetero Labs and Hetero USA should be deemed submitters of ANDA No. 221270 together with Annora.

JURISDICTION AND VENUE

11. This Court has subject matter jurisdiction pursuant to 28 U.S.C. §§ 1331, 1338(a), 2201, and 2202.

12. This Court has personal jurisdiction over Defendant Hetero USA because, on information and belief, Hetero USA is a corporation organized and existing under the laws of the State of Delaware.

13. This Court has personal jurisdiction over Defendants because, *inter alia*, Defendants, themselves and through their subsidiaries, agents, and/or affiliates, have purposefully availed themselves of the benefits and protections of Delaware's laws such that they should reasonably anticipate being haled into court here. Upon information and belief, Defendants, themselves and through their subsidiaries, agents, and/or affiliates, develop, manufacture, import, market, offer to sell, sell, and/or distribute a broad range of generic pharmaceutical products throughout the United States, including in Delaware, and therefore transact business within Delaware relating to Plaintiff's claims, and/or have engaged in systematic and continuous business contacts within Delaware.

14. In addition, this Court has personal jurisdiction over Defendants because, among other things, upon information and belief: (1) Defendants acted in concert with respect to the development, preparation, and filing of Annora's Proposed ANDA product that is the subject of ANDA No. 221270 and submitted ANDA No. 221270 for the purpose of seeking approval to engage in the commercial manufacture, use, sale, or offer for sale in the United States of Annora's Proposed ANDA Product, including in Delaware; (2) upon approval of Annora's ANDA, Defendants will act in concert to market, distribute, offer for sale, and sell in the United States, and/or import into the United States, including in Delaware, Annora's Proposed ANDA Product, and will derive substantial revenue from the use or consumption of Annora's Proposed ANDA Product in Delaware; (3) upon approval of Annora's ANDA, Annora's Proposed ANDA Product will, among other things, be marketed, distributed, offered for sale, and sold in Delaware, and/or imported into Delaware; prescribed by physicians practicing in Delaware and administered to patients in Delaware, all of which would have substantial effects on Delaware. By submitting Annora's ANDA, Defendants have shown they will use distribution channels to direct sales of Annora's Proposed ANDA Product into Delaware.

15. In addition, this Court has personal jurisdiction over Defendants because they have committed, aided, abetted, induced, contributed to, or participated in the commission of the tortious act of patent infringement that has led and/or will lead to foreseeable harm and injury to Agios, a Delaware corporation.

16. Furthermore, this Court has personal jurisdiction over Defendants because they have regularly engaged in patent litigation concerning Annora's ANDA products in this District and have consented to jurisdiction in Delaware in prior cases arising out of the filing of ANDAs. *Sumitomo Pharma Switzerland GmbH, et al., v. Annora Pharma Private Limited, et al.*, 25-380-MN (D. Del.), *Mirum Pharmaceuticals, Inc., et al., v. Annora Pharma Private*

Limited, et al., No. 25-1537-MN (D. Del.), *Sumitomo Pharma Switzerland GmbH, et al., v. Annora Pharma Private Limited, et al.*, C.A. No. 25-380-MN (D. Del.), *Azurity Pharmaceuticals, Inc., et al., v. Annora Pharma Private Ltd.*, C.A. No. 25-1493-MN (D. Del.), *Azurity Pharmaceuticals, Inc. v. Annora Pharma Private Ltd.*, C.A. No. 25-1139-CFC (D. Del.).

17. In the alternative, this Court has personal jurisdiction over Annora and Hetero Labs under Federal Rule of Civil Procedure 4(k)(2) because (a) Plaintiff's claims arise under federal law; (b) Annora and Hetero Labs are foreign companies not subject to jurisdiction in any state's courts of general jurisdiction; and (c) Annora and Hetero Labs have sufficient contacts with the United States as a whole, including but not limited to marketing and/or selling generic pharmaceutical products that are distributed and sold throughout the United States, such that this Court's exercise of jurisdiction over Annora and Hetero Labs satisfies due process.

18. Venue is proper in this judicial district as to Annora and Hetero Labs under 28 U.S.C. § 1391(c)(3) at least because, upon information and belief, Annora and Hetero Labs are foreign corporations that may be sued in any judicial district in the United States.

19. Venue is proper in this judicial district as to Hetero USA under 28 U.S.C. § 1400(b) because, *inter alia*, Hetero US is incorporated in Delaware and therefore resides in Delaware.

BACKGROUND

PYRUKYND® (MITAPIVAT)

20. Pyrukynd® is a pyruvate kinase activator indicated for the treatment of hemolytic anemia in adults with pyruvate kinase (PK) deficiency.

21. Agios sells Pyrukynd® in the United States pursuant to New Drug Application ("NDA") No. 216196, which has been approved by the FDA.

22. Agios is the holder of approved NDA No. 216196 for Pyrukynd®.

23. The Asserted Patents are listed for NDA No. 216196 for Pyrukynd® in the FDA's *Approved Drug Products with Therapeutic Equivalence Evaluations*, commonly known as the "Orange Book."

24. The '976 Patent, titled "Methods of Using Pyruvate Kinase Activators," was duly and legally issued on February 1, 2022. A copy of the '976 Patent is attached as Exhibit A.

25. Agios is the assignee of the '976 Patent.

26. There is an actual case or controversy between the parties regarding Defendants' liability for their infringement of the '976 Patent.

27. The '652 Patent, titled "Crystalline Forms of N-(4-(4-(Cyclopropylmethyl)piperazine-1-carbonyl)phenyl)quinoline-8-sulfonamide," was duly and legally issued on February 22, 2022. A copy of the '652 Patent is attached as Exhibit B.

28. Agios is the assignee of the '652 Patent.

29. There is an actual case or controversy between the parties regarding Defendants' liability for their infringement of the '652 Patent.

30. The ’049 Patent, titled “Mitapivat Therapy and Modulators of Cytochrome P450,” was duly and legally issued on January 23, 2024. A copy of the ’049 Patent is attached as Exhibit C.

31. Agios is the assignee of the ’049 Patent.

32. There is an actual case or controversy between the parties regarding Defendants’ liability for their infringement of the ’049 Patent.

33. This action is being filed within 45 days of Plaintiff’s receipt of Annora’s Notice Letter.

ANNORA’S ANDA

34. According to Annora’s Notice Letter, ANDA No. 221270 includes a certification pursuant to the Food, Drug, & Cosmetic Act (“FDCA”) 21 U.S.C. § 355(j)(2)(A)(vii)(IV) (“Paragraph IV Certification”) that the Asserted Patents are invalid, unenforceable, and/or will not be infringed by the commercial manufacture, use, sale, or offer for sale of Annora’s Proposed ANDA Product.

35. Defendants were aware of the Asserted Patents when ANDA No. 221270 was filed with a Paragraph IV Certification.

36. Upon information and belief, mitapivat, present as mitapivat sulfate, is the active ingredient in Annora’s Proposed ANDA Product.

37. Upon information and belief, ANDA No. 221270 refers to and relies upon the NDA for Pyrukynd® and contains data that, according to Defendants, demonstrate bioequivalence of Annora’s Proposed ANDA Product and Pyrukynd®, see 21 U.S.C. § 355(j)(2); 21 C.F.R. § 314.94(a)(7), or Defendants have sought a waiver of the requirement to demonstrate bioequivalence of their Proposed ANDA Product and Pyrukynd®.

38. Upon information and belief, Defendants intend to have healthcare professionals administer Annora’s Proposed ANDA Product, if approved, as set forth in Annora’s proposed label for this ANDA product. Upon information and belief, Annora’s proposed label will instruct healthcare professionals to prescribe and administer Annora’s Proposed ANDA Product in the manner set forth in the label.

COUNT I

(Infringement of the ’976 Patent)

39. Plaintiff incorporates each of the preceding paragraphs as if fully set forth herein.

40. Claim 1 of the ’976 Patent covers “A method for treating pyruvate kinase deficiency (PKD) in a subject in need thereof, comprising orally administering to the subject once or twice daily a dose of about 10 mg to about 60 mg or about 60 mg to about 200 mg of (1) Compound 1 or a pharmaceutically acceptable salt or hydrate thereof; or (2) a pharmaceutical composition comprising Compound 1 or a pharmaceutically acceptable salt or hydrate thereof, and a pharmaceutically acceptable carrier, to thereby treat PKD in the subject, wherein Compound 1 is N-(4-(4-(cyclopropylmethyl)piperazine-1-carbonyl)phenyl)quinoline-8-sulfonamide.”

41. Upon information and belief, Annora’s Proposed ANDA Product and its use are covered by one or more of the claims of the ’976 Patent, including at least claim 1.

42. Upon information and belief, the commercial manufacture, sale, offer for sale, and/or importation of Annora's Proposed ANDA Product, and/or the use of Annora's Proposed ANDA Product in accordance with and as directed by Annora's proposed label for that product, will infringe one or more claims of the '976 Patent, including at least claim 1, either literally or under the doctrine of equivalents.

43. Upon information and belief, Defendants submitted as part of ANDA No. 221270 a certification of the type described in Section 505(j)(2)(A)(vii)(IV) of the FDCA, 21 U.S.C. § 355(j)(2)(A)(vii)(IV), asserting that the claims of the '976 Patent are invalid, unenforceable, and/or not infringed by the commercial manufacture, use, offer for sale, or sale of Annora's Proposed ANDA Product.

44. Defendants did not contend in Annora's Notice Letter that Annora's Proposed ANDA Product, or the use of Annora's Proposed ANDA Product in accordance with and as directed by Annora's proposed label for that product, would not infringe the '976 Patent.

45. The purpose of submitting ANDA No. 221270 was to obtain approval under the FDCA to engage in the commercial manufacture, use, offer for sale, and/or sale of Annora's Proposed ANDA Product prior to the expiration of the '976 Patent.

46. Defendants' submission of ANDA No. 221270 for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, and/or sale of Annora's Proposed ANDA Product prior to the expiration of the '976 Patent was an act of infringement of the '976 Patent under 35 U.S.C. § 271(e)(2)(A).

47. Upon information and belief, Defendants intend to engage in the commercial manufacture, use, offer for sale, sale, marketing, distribution, and/or importation of Annora's Proposed ANDA Product and the proposed label immediately and imminently upon the approval of ANDA No. 221270 and any amendments thereto, i.e., prior to the expiration of the '976 Patent, in violation of 35 U.S.C. § 271(a).

48. Upon information and belief, Defendants have knowledge of the '976 Patent at least because the '976 Patent is listed in the FDA's Orange Book for Plaintiff's Pyrukynd® drug product. Notwithstanding this knowledge, Defendants continue to assert their intent to engage in the commercial manufacture, use, offer for sale, sale, marketing, distribution, and/or importation of Annora's Proposed ANDA Product and the proposed label therefor immediately and imminently upon the approval of ANDA No. 221270.

49. Upon information and belief, Defendants plan and intend to, and will, actively induce infringement of the '976 Patent when ANDA No. 221270 is approved, and will do so with specific intent to induce infringement of the '976 Patent, in violation of 35 U.S.C. § 271(b). Further upon information and belief, Defendants plan and intend to, and will, do so immediately and imminently upon approval.

50. Upon information and belief, Defendants know that Annora's Proposed ANDA Product is especially made or adapted for use in patented methods of the '976 Patent, and that Annora's Proposed ANDA Product is not suitable for substantial noninfringing use. Upon information and belief, Defendants plan and intend to, and will, contribute to infringement of the '976

Patent immediately and imminently upon approval of ANDA No. 221270, in violation of 35 U.S.C. 271(c).

51. The foregoing actions by Defendants constitute and/or will constitute infringement of the ’976 Patent, active inducement of infringement of the ’976 Patent, and contribution to the infringement by others of the ’976 Patent, either literally or under the doctrine of equivalents.

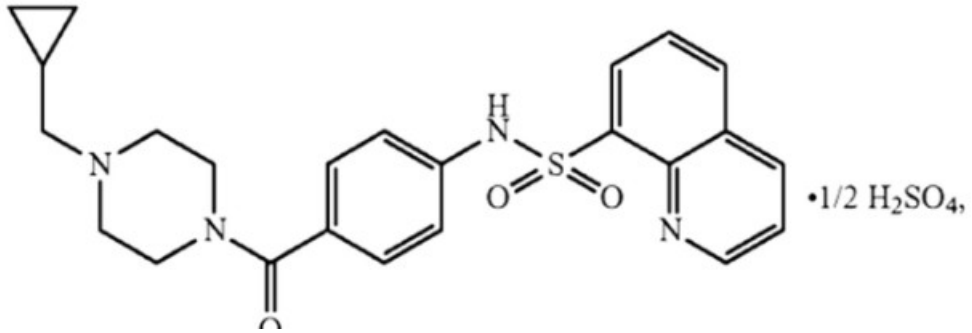
52. Unless Defendants are enjoined from infringing the ’976 Patent, actively inducing infringement of the ’976 Patent, and contributing to the infringement by others of the ’976 Patent, Agios will suffer irreparable injury. Agios has no adequate remedy at law.

COUNT II

(Infringement of the ’652 Patent)

53. Plaintiff incorporates each of the preceding paragraphs as if fully set forth herein.

54. Claim 1 of the ’652 Patent covers “A crystalline form of a compound having the formula:



wherein the crystalline form is crystalline form A characterized by x-ray powder diffraction peaks at 2Θ angles (±0.2°) 9.9°, 15.8°, and 22.6°.”

55. Upon information and belief, Annora’s Proposed ANDA Product and its use are covered by one or more of the claims of the ’652 Patent, including at least claim 1.

56. Upon information and belief, the commercial manufacture, sale, offer for sale, and/or importation of Annora’s Proposed ANDA Product, and/or the use of Annora’s Proposed ANDA Product in accordance with and as directed by Annora’s proposed label for that product, will infringe one or more claims of the ’652 Patent, including at least claim 1, either literally or under the doctrine of equivalents.

57. Upon information and belief, Defendants submitted as part of ANDA No. 221270 a certification of the type described in Section 505(j)(2)(A)(vii)(IV) of the FDCA, 21 U.S.C. § 355(j)(2)(A)(vii)(IV), asserting that the claims of the ’652 Patent are invalid, unenforceable, and/or not infringed by the commercial manufacture, use, offer for sale, or sale of Annora’s Proposed ANDA Product.

58. The purpose of submitting ANDA No. 221270 was to obtain approval under the FDCA to engage in the commercial manufacture, use, offer for sale, and/or sale of Annora’s Proposed ANDA Product prior to the expiration of the ’652 Patent.

58. Defendants' submission of ANDA No. 221270 for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, and/or sale of Annora's Proposed ANDA Product prior to the expiration of the '652 Patent was an act of infringement of the '652 Patent under 35 U.S.C. § 271(e)(2)(A).

59. Upon information and belief, Defendants intend to engage in the commercial manufacture, use, offer for sale, sale, marketing, distribution, and/or importation of Annora's Proposed ANDA Product and the proposed label immediately and imminently upon the approval of ANDA No. 221270 and any amendments thereto, i.e., prior to the expiration of the '652 Patent, in violation of 35 U.S.C. § 271(a).

60. Upon information and belief, Defendants have knowledge of the '652 Patent at least because the '652 Patent is listed in the FDA's Orange Book for Plaintiff's Pyrukynd® drug product. Notwithstanding this knowledge, Defendants continue to assert their intent to engage in the commercial manufacture, use, offer for sale, sale, marketing, distribution, and/or importation of Annora's Proposed ANDA Product and the proposed label therefor immediately and imminently upon the approval of ANDA No. 221270.

61. Upon information and belief, Defendants plan and intend to, and will, actively induce infringement of the '652 Patent when ANDA No. 221270 is approved, and will do so with specific intent to induce infringement of the '652 Patent, in violation of 35 U.S.C. § 271(b). Further upon information and belief, Defendants plan and intend to, and will, do so immediately and imminently upon approval.

62. Upon information and belief, Defendants know that Annora's Proposed ANDA Product is especially made or adapted for use in patented methods of the '652 Patent, and that Annora's Proposed ANDA Product is not suitable for substantial noninfringing use. Upon information and belief, Defendants plan and intend to, and will, contribute to infringement of the '652 Patent immediately and imminently upon approval of ANDA No. 221270, in violation of 35 U.S.C. § 271(c).

63. The foregoing actions by Defendants constitute and/or will constitute infringement of the '652 Patent, active inducement of infringement of the '652 Patent, and contribution to the infringement by others of the '652 Patent, either literally or under the doctrine of equivalents.

64. Unless Defendants are enjoined from infringing the '652 Patent, actively inducing infringement of the '652 Patent, and contributing to the infringement by others of the '652 Patent, Agios will suffer irreparable injury. Agios has no adequate remedy at law.

COUNT III

(Infringement of the '049 Patent)

66. Plaintiff incorporates each of the preceding paragraphs as if fully set forth herein.

67. Claim 1 of the '049 Patent covers "A method of treating hemolytic anemia in a subject, comprising administering to the subject about 10 mg/day, about 40 mg/day, about 100 mg/day, or about 200 mg/day of mitapivat or a pharmaceutically acceptable salt thereof and a moderate inducer of cytochrome P450 3A4/5 (CYP3A4/5)."

68. Upon information and belief, Annora's Proposed ANDA Product and its use are covered by one or more of the claims of the '049 Patent, including at least claim 1.

69. Upon information and belief, the commercial manufacture, sale, offer for sale, and/or importation of Annora's Proposed ANDA Product, and/or the use of Annora's Proposed ANDA Product in accordance with and as directed by Annora's proposed label for that product, will infringe one or more claims of the '049 Patent, including at least claim 1, either literally or under the doctrine of equivalents.

70. Upon information and belief, Defendants submitted as part of ANDA No. 221270 a certification of the type described in Section 505(j)(2)(A)(vii)(IV) of the FDCA, 21 U.S.C. § 355(j)(2)(A)(vii)(IV), asserting that the claims of the '049 Patent are invalid, unenforceable, and/or not infringed by the commercial manufacture, use, offer for sale, or sale of Annora's Proposed ANDA Product.

71. The purpose of submitting ANDA No. 221270 was to obtain approval under the FDCA to engage in the commercial manufacture, use, offer for sale, and/or sale of Annora's Proposed ANDA Product prior to the expiration of the '049 Patent.

72. Defendants' submission of ANDA No. 221270 for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, and/or sale of Annora's Proposed ANDA Product prior to the expiration of the '049 Patent was an act of infringement of the '049 Patent under 35 U.S.C. § 271(e)(2)(A).

73. Upon information and belief, Defendants intend to engage in the commercial manufacture, use, offer for sale, sale, marketing, distribution, and/or importation of Annora's Proposed ANDA Product and the proposed label immediately and imminently upon the approval of ANDA No. 221270 and any amendments thereto, i.e., prior to the expiration of the '049 Patent, in violation of 35 U.S.C. § 271(a).

74. Upon information and belief, Defendants have knowledge of the '049 Patent at least because the '049 Patent is listed in the FDA's Orange Book for Plaintiff's Pyrukynd® drug product. Notwithstanding this knowledge, Defendants continue to assert their intent to engage in the commercial manufacture, use, offer for sale, sale, marketing, distribution, and/or importation of Annora's Proposed ANDA Product and the proposed label therefor immediately and imminently upon the approval of ANDA No. 221270.

75. Upon information and belief, Defendants plan and intend to, and will, actively induce infringement of the '049 Patent when ANDA No. 221270 is approved, and will do so with specific intent to induce infringement of the '049 Patent, in violation of 35 U.S.C. § 271(b). Further upon information and belief, Defendants plan and intend to, and will, do so immediately and imminently upon approval.

76. Upon information and belief, Defendants know that Annora's Proposed ANDA Product is especially made or adapted for use in patented methods of the '049 Patent, and that Annora's Proposed ANDA Product is not suitable for substantial noninfringing use. Upon information and belief, Defendants plan and intend to, and will, contribute to infringement of the '049

Patent immediately and imminently upon approval of ANDA No. 221270, in violation of 35 U.S.C. § 271(c).

77. The foregoing actions by Defendants constitute and/or will constitute infringement of the ’049 Patent, active inducement of infringement of the ’049 Patent, and contribution to the infringement by others of the ’049 Patent, either literally or under the doctrine of equivalents.

78. Unless Defendants are enjoined from infringing the ’049 Patent, actively inducing infringement of the ’049 Patent, and contributing to the infringement by others of the ’049 Patent, Agios will suffer irreparable injury. Agios has no adequate remedy at law.

REQUEST FOR RELIEF

WHEREFORE, Plaintiff requests the following relief:

(a) A judgment that Defendants have infringed the Asserted Patents under 35 U.S.C. § 271(e)(2)(A);

(b) A judgment pursuant to 35 U.S.C. § 271(e)(4)(A) ordering that the effective date of any FDA approval for Defendants to make, use, offer for sale, sell, market, distribute, or import Annora’s Proposed ANDA Product, or any product the use of which infringes the Asserted Patents, be not earlier than the latest expiration date among the Asserted Patents, inclusive of any extension(s) and additional period(s) of exclusivity;

(c) A permanent injunction pursuant to 35 U.S.C. § 271(e)(4)(B) enjoining Defendants, and all persons and entities acting in concert with Defendants, from making, using, selling, offering for sale, marketing, distributing, or importing Annora’s Proposed ANDA Product, or any product the use of which infringes the Asserted Patents, or the inducement of or contribution to any of the foregoing, prior to the expiration date of the latest-expiring of the Asserted Patents, inclusive of any extension(s) and additional period(s) of exclusivity;

(d) A judgment declaring that making, using, selling, offering for sale, marketing, distributing, or importing of Annora’s Proposed ANDA Product, or any product the use of which infringes the Asserted Patents, prior to the expiration date of the latest-expiring of the Asserted Patents, infringes, will infringe, will actively induce infringement of, and/or will contribute to the infringement by others of the Asserted Patents under 35 U.S.C. §§ 271(a), (b) and (c);

(e) An award of damages or other relief, pursuant to 35 U.S.C. § 271(e)(4)(C), if Defendants engage in making, using, selling, offering for sale, marketing, distributing, or importing of Annora’s Proposed ANDA Product, or any product the use of which infringes the Asserted Patents, or induces or contributes to such conduct, prior to the expiration date of the latest-expiring of the Asserted Patents, inclusive of any extension(s) and additional period(s) of exclusivity;

(f) A finding that this is an exceptional case and an award of attorneys’ fees pursuant to 35 U.S.C. § 285;

(g) An award of Plaintiff’s costs and expenses in this action; and

(h) Such further and other relief as this Court may deem just and proper.

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